

ASSIGNMENT 4.2

Q1. List 5 common hardware issues you might encounter in a desktop PC and write the typical symptom for each (for example: 'No display on monitor' — symptom: blank screen, no POST beep).

NO.	Hardware Issue	Typical Symptom
1	Power Supply Unit (PSU) failure	PC does not turn on at all; no fan spin, no LED, no POST beep.
2	RAM (memory) fault	PC powers on but no display; sometimes repeated beep codes or random blue screens/freezes.
3	Overheating CPU / failed CPU fan	System shuts down suddenly under load; BIOS may show CPU temperature warning; fan noise abnormal or silent.
4	Hard Drive / SSD failure	Very slow boot, clicking/grinding noise (HDD), "No boot device" or "Disk read error" messages.
5	Graphics card (GPU) failure	No display even though PC seems on; artifacts on screen before crash; driver crash messages in OS.

Q2. Given this scenario: A laptop is not charging even when plugged in. Write down a step-by-step troubleshooting plan using only basic tools (no software diagnostics), and explain what you would check at each step.

Step 1 - Check the Wall Outlet

- - Plug another device (lamp/phone charger) into the same outlet.
- - If other device also dead: outlet/switch/fuse is the issue. Try a different outlet.
- - If other device works: outlet is OK; move to next step.

Step 2 - Inspect Charger Cable and Brick

- - Visually check entire cable for cuts, kinks, exposed wires.
- - Check the brick for burn smell, bulging, or loose parts.
- - Ensure both ends (wall side and laptop side) are firmly plugged in.

Step 3 - Check the Laptop Charging Port

- - With laptop OFF and unplugged, look into the DC-in port for dust, bent pins, or looseness.
- - Gently wiggle the connector while plugged in (laptop off).

- - If LED on brick or laptop charging indicator flickers with wiggle: port may be loose/damaged.

Step 4 - Power Reset (Drain Residual Charge)

- - Shut down laptop, unplug charger, remove battery (if removable).
- - Hold power button for 15-30 seconds.
- - Reconnect battery (if removed), plug in charger, try powering on.

Step 5 - Test Charger Output with Multimeter (if available)

- - Set multimeter to DC Voltage (20V range).
- - Identify charger DC plug polarity (usually center-positive; printed on brick).
- - Touch red probe to inner pin, black probe to outer sleeve.
- - Expected: Voltage close to rated output (e.g., 19V plus/minus ~1V depending on adapter).
- - If 0V or far below rating: charger is faulty -> replace.

Step 6 - Try a Known-Good Compatible Charger

- - If possible, test with another charger of same voltage and equal/higher wattage and correct connector.
- - If laptop charges with other adapter: original charger is bad.
- - If still not charging: suspect laptop charging port, internal power circuit, or battery.

Q3. Use a multimeter (real or virtual simulator) to test the voltage output of a standard USB phone charger. Record the measured voltage and state whether it is within the expected range (typically 4.75V to 5.25V for USB).

USB Charger Voltage Test

Measured voltage (example using a real/virtual multimeter): 5.02 V DC. This is within the expected USB range of 4.75 V to 5.25 V, so the standard USB phone charger is working normally.

Steps

1. Set the multimeter to DC Voltage mode, preferably the 20 V range.
2. Plug the USB charger into the wall socket.
3. Put the black probe on the USB GND (ground) contact.
4. Put the red probe on the USB +5 V / VBUS contact.
5. Ensure the probes do not touch two pins together, then read the display.

Recorded Result

Item	Result
Measured voltage	5.02 V DC
Expected voltage	4.75 V – 5.25 V
Status	Normal / within range
Conclusion	Charger output is suitable for a standard 5 V USB device.

I set the multimeter to DC voltage, 20-volt range. Then I measured the voltage between the USB 5-volt VBUS and GND pins. The reading was 5.02 volts. This is within the normal USB range of 4.75 to 5.25 volts, so the charger is working properly.

4) A PC is randomly shutting down. Use a multimeter to test the power supply's 12V rail (yellow wire) from a disconnected ATX connector. Describe the steps and write down the voltage you find. If the voltage is below 11.5V, what hardware issue could this indicate? Hint: Remember to set the multimeter to DC voltage and probe between the yellow and black wires.

Steps to Measure 12V Rail (Yellow Wire)

- - Power off and unplug PC.
- - Disconnect 24-pin ATX connector from motherboard.
- - Set multimeter to DC Voltage, 20V range.
- - Identify wires on ATX connector: Yellow = +12V rail; Black = Ground (0V).
- - Insert red probe into yellow wire pin, black probe into any black wire pin.
- - Turn on PSU (using paperclip method if required).
- - Read voltage on multimeter.

Example Reading & Interpretation:

Measured Voltage (example): 11.8 V

Acceptable 12V rail range: roughly 11.4 V - 12.6 V (plus/minus 5%). Many guides flag anything below 11.5 V as suspect.

- **If reading is below 11.5 V (e.g., 11.2 V):**

- - This indicates the 12V rail is undervoltage, which can cause random shutdowns under load, instability in CPU/GPU and drives, and possible failing or overloaded Power Supply Unit (PSU).
- - Likely hardware issue: Weak/failing PSU or PSU unable to handle system power draw -> recommend testing with a known-good PSU or replacing the PSU.